UNIVERSITETET I OSLO

IN5340 / IN9340 Lecture 4
Parameter Estimation II

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Contents

What do we learn

- Parameter estimation part II
- Ch 6 in Therrien and the whole book of Kay

S. M. Kay.
Fundamentals of Statistical Signal Processing: Estimation Theory, Volume I.
Prentice Hall International Editions, 1993.

C. W. Therrien.
Discrete Random Signals and Statistical Signal Processing.
Prentice Hall, 1992.

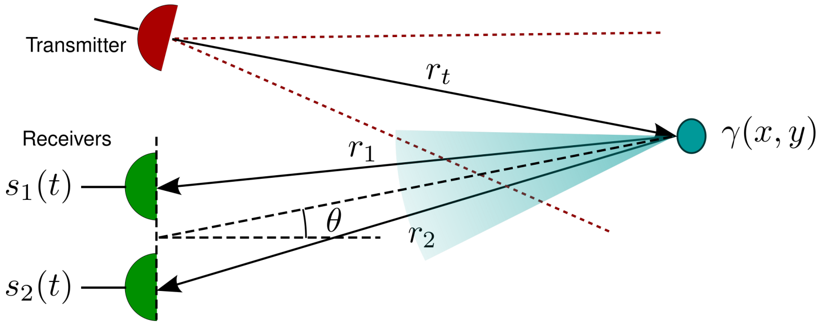
Additional information

- The course `EE 522` by Prof. Mark Fowler at Binghamton University has excellent lecture notes
- "Google" for EE522 and download / read:
  - `Notes #4 Ch 3 Cramer Rao Bound Pt. A`
  - `Notes #5 Ch 3 Cramer Rao Bound Pt. B`
  - `Notes #8 Ch 3 CRLB Examples`
  - `Notes #16 Ch 8A LSE`
- Note 4 (5,6,7) introduces the Cramér-Rao lower bound
- Note 8 has the example of CRLB for time delay estimation
- Note 16 (17,18,19) covers Least Squares Estimation
- Least Squares Estimation is excellent covered in wikipedia

`https://ws.binghamton.edu/fowler/fowler%20personal%20page/EE522.htm`

Motivation

- Example: Bearing estimation
- The task: Find an optimal estimator and its performance



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Parameter estimation

- *Parameter estimation* is to estimate a certain parameter from a (random) sequence
- This can be mathematically formulated by the probability density function, where the data is dependent on the unknown parameter

  $f_x(\mathbf{x}; \theta)$
- As a function of the parameter, this is called the *likelihood function*
- The performance of any estimator can be characterised by estimator properties such as *bias* and *variance*

Parameter estimation grand goal

Finding an optimal estimator and assessing its performance

Optimal estimators

- There are a number of different optimisation criteria
- The optimal estimator does not always exsist...
- Maximizing the likelihood function leads to the *maximum likelihood estimator*
- Choosing the unbiased estimator with minimum variance leads the *Minimum Variance Unbiased* estimator

Parameter estimation example

- typical model: signal with unknown parameters in noise
  $x[n] = s[n, \theta] + w[n]$
- Example: a constant unknown value A in AWGN:
  $x[n] = A + w[n]$
- The PDF (or likelihood function) becomes
  $$f_x(\mathbf{x}; \theta) = \frac{1}{(2\pi\sigma_x^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_x^2}\sum_{n=1}^{N}(x[n] - A)^2\right)$$
- We remember that the sample average
  $$\hat{A} = \frac{1}{N}\sum_{n=1}^{N} x[n]$$
  is an unbiased estimator that is consistent.
  It is also the maximum likelihood estimator

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Cramér-Rao lower bound

Cramér-Rao lower bound

The *Cramér-Rao lower bound* (CRLB) is the theoretical lower bound on the *variance* of any *unbiased* estimator

- If $\hat{\theta}$ is an unbiased estimator of θ
  $$\mathrm{Var}(\hat{\theta}) \geq \mathrm{CRLB}(\theta) \qquad \text{for all estimators } \hat{\theta}$$
- Note that the CRLB is only a function of the mathematical model.
- The CRLB is not a function of the estimator
- The CRLB states the best possible performance any estimator can have.

Harald Cramér (1893 – 1985). Swedish mathematician and statistician. From wikipedia.

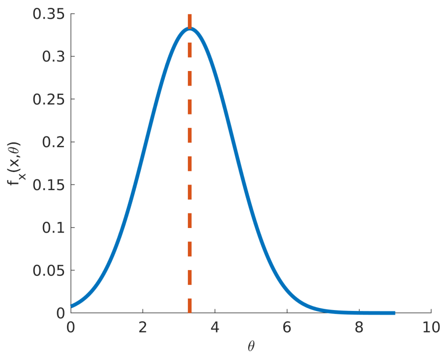
Calyampudi Radhakrishna Rao (1920 – 2023). Indian-American mathematician and statistician. Photo: Prateek Karandikar. From wikipedia.

Applications of the Cramér-Rao lower bound

- Modeling of sensor performance
- Benchmarking of new estimators
  - Simulate a given scenario with known θ
  - Design and implement new estimator
  - Compare measured error with CRLB
- Feasibility studies:
  - Can some sensor actually meet the specifications
  - Has the system been oversold
- Assess estimator performance - how close are we
- Can be used to check if an estimator is the MVU estimator

Intuitive description of Cramér-Rao lower bound

- Recall that the *maximum likelihood* estimate was found by finding
$$\left.\frac{\partial \ln f_x(\mathbf{x},\theta)}{\partial\theta}\right|_{\theta=\hat{\theta}_{ml}} = 0$$
- The sharpness of the likelihood function sets the accuracy
- The sharpness can be measured using curvature:
$$-\left.\frac{\partial^2 \ln f_x(\mathbf{x},\theta)}{\partial\theta^2}\right|_{\substack{\mathbf{x}=\text{given data}\\ \theta=\text{true value}}}$$



Mathematical description of Cramér-Rao lower bound

Theorem

- Assume that the *regularity condition* is met
$$E\left\{\frac{\partial \ln f_x(\mathbf{x},\theta)}{\partial\theta}\right\} = 0 \qquad \text{for all } \theta$$
where $E\{\cdot\}$ is with respect to the likelihood function
- Generally true except when the domain of the PDF for which it is nonzero depends on the unknown parameter θ

Mathematical description of Cramér-Rao lower bound

Theorem cont.

- The variance for any unbiased estimator must satisfy
$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{-E\left\{\frac{\partial^2 \ln f_x(\mathbf{x},\theta)}{\partial\theta^2}\right\}}$$
where the derivative is taken at the true value of θ
- The expectation is taken with respect to $f_x(\mathbf{x},\theta)$

Cookbook for finding the Cramér-Rao lower bound

1. Write the log likelihood function (LLF) (the PDF as a function θ)
$$\ln f_x(\mathbf{x},\theta)$$
2. Take the second derivative of the LLF
$$\frac{\partial^2 \ln f_x(\mathbf{x},\theta)}{\partial\theta^2}$$
3. If still a function of $\mathbf{x}$, take expectation with respect to $\mathbf{x}$
4. If still a function of θ, evaluate at desired value
5. Negate and invert

Example: CRLB for estimating a constant in AWGN

- Example: Constant unknown value A in AWGN: $x[n] = A + w[n]$
- The likelihood function is
$$f_x(\mathbf{x};\theta) = \frac{1}{(2\pi\sigma_x^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_x^2}\sum_{n=1}^{N}(x[n]-A)^2\right)$$
- The log likelihood function becomes
$$\ln f_x(\mathbf{x},A) = -\frac{N}{2}\ln(2\pi\sigma_x^2) - \frac{1}{2\sigma_x^2}\sum_{n=1}^{N}(x[n]-A)^2$$
- The first derivative of the LLF
$$\frac{\partial \ln f_x(\mathbf{x},A)}{\partial A} = \frac{N}{\sigma_x^2}\underbrace{\frac{1}{N}\sum_{n=1}^{N}x[n]}_{\text{sample mean } \bar{x}} - \frac{A}{\sigma_x^2}N$$

Example: CRLB for estimating a constant in AWGN

- The second derivative of the LLF becomes
$$\frac{\partial}{\partial A}\left\{\frac{N}{\sigma_x^2}(\bar{x}-A)\right\} = -\frac{N}{\sigma_x^2}$$
- Final stage: negate and invert:
$$\mathrm{CRLB} = \frac{\sigma_x^2}{N}$$
- The variance for all unbiased estimators of a constant value in AWGN must fulfill
$$\mathrm{var}(\hat{A}) \geq \frac{\sigma_x^2}{N}$$
- The CRLB for estimating a constant value in AWGN decreases with increased observation set, and increases with increasing noise standard deviation

Mathematical description of Cramér-Rao lower bound

Continuation of the theorem

- There exists an unbiased estimator that attains the Cramér-Rao lower bound if and only if
$$\frac{\partial \ln f_x(\mathbf{x},\theta)}{\partial\theta} = I(\theta)(g(\mathbf{x})-\theta)$$
for some function g and I
- That estimator, which is the MVU estimator is
$$\hat{\theta} = g(\mathbf{x})$$
- The minimum variance is
$$\mathrm{CRLB}(\theta) = \frac{1}{I(\theta)}$$
- $I(\theta)$ is called the *Fisher Information*

Ronald A. Fisher 1890 - 1962
British statistician and geneticist
From Wikipedia

Example: A constant value in AWGN revisited

- Example: Our signal in noise $x[n] = A + w[n]$
- The likelihood function is
$$f_x(\mathbf{x},A) = \frac{1}{(2\pi\sigma_x^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_x^2}\sum_{n=1}^{N}(x[n]-A)^2\right)$$
- The log likelihood function was found to be
$$\ln f_x(\mathbf{x},A) = -\frac{N}{2}\ln(2\pi\sigma_x^2) - \frac{1}{2\sigma_x^2}\sum_{n=1}^{N}(x[n]-A)^2$$
- And the first derivative of the log likelihood function was
$$\frac{\partial \ln f_x(\mathbf{x},A)}{\partial A} = \frac{N}{\sigma_x^2}(\bar{x}-A)$$

Example: A constant value in AWGN revisited

- Continuing...

$$\frac{\partial \ln f_x(\mathbf{x}, A)}{\partial A} = \frac{N}{\sigma_x^2}(\bar{x} - A)$$

- We immediately recognise the right hand side as

$$I(\theta)(g(\mathbf{x}) - \theta)$$

where

$$\hat{\theta} = \bar{x}$$

is the MVU estimator, and

$$\mathrm{CRLB}(\theta) = \frac{1}{I(\theta)} = \frac{\sigma_x^2}{N}$$

Alternative form of Cramér-Rao lower bound

Alternative forms of the Cramér-Rao lower bound

- Recall that

$$\mathrm{CRLB}(\theta) = \frac{1}{I(\theta)}$$

- $I(\theta)$ is called the *Fisher Information*
- Can be represented in two equivalent forms:

$$I(\theta) = -E\left\{\frac{\partial^2 \ln f_x(\mathbf{x}, \theta)}{\partial \theta^2}\right\} = \underbrace{E\left\{\left(\frac{\partial \ln f_x(\mathbf{x}, \theta)}{\partial \theta}\right)^2\right\}}_{\text{alternative form}}$$

- See appendix 3A in [Kay(1993)] for the proof (uses the regularity condition)

Alternative form of CRLB – Calculation (1)

- OK. Let's try calculating it. Here we go:
- The regularity condition

$$E\left\{\frac{\partial \ln f_x(\mathbf{x}, \theta)}{\partial \theta}\right\} = 0$$

- Remember: the expectation of any function

$$E\{g(x)\} = \int g(\alpha) f_x(\alpha) d\alpha$$

- Hence

$$\int \frac{\partial \ln f_x(\mathbf{x}, \theta)}{\partial \theta} f_x(\mathbf{x}, \theta) = 0 \quad \Rightarrow \quad \frac{\partial}{\partial \theta} \int \frac{\partial \ln f_x(\mathbf{x}, \theta)}{\partial \theta} f_x(\mathbf{x}, \theta) = 0$$

Alternative form of CRLB – Calculation (2)

- The product rule $(uv)' = u'v + uv'$

$$\int \frac{\partial^2 \ln f_x(\mathbf{x}, \theta)}{\partial \theta^2} f_x(\mathbf{x}, \theta) + \int \frac{\partial \ln f_x(\mathbf{x}, \theta)}{\partial \theta} \frac{\partial f_x(\mathbf{x}, \theta)}{\partial \theta} = 0$$

- Now

$$\frac{d}{dx} \ln f(x) = \frac{1}{f(x)} \frac{d}{dx} f(x) \quad \Rightarrow \quad \frac{d}{dx} f(x) = f(x) \frac{d}{dx} \ln f(x)$$

- which gives

$$E\left\{\frac{\partial^2 \ln f_x(\mathbf{x}, \theta)}{\partial \theta^2}\right\} = -E\left\{\left(\frac{\partial \ln f_x(\mathbf{x}, \theta)}{\partial \theta}\right)^2\right\}$$

- q.e.d.

CRLB for a constant value in AWGN - again

- Our simple example. We recall (after regrouping)

$$\frac{\partial \ln f_x(\mathbf{x}, A)}{\partial A} = \frac{1}{\sigma_x^2} \sum_{n=1}^{N} (x[n] - A)$$

- The alternative form is

$$I(\theta) = E\left\{ \left(\frac{\partial \ln f_x(\mathbf{x}, \theta)}{\partial \theta} \right)^2 \right\} = E\left\{ \left(\frac{1}{\sigma_x^2} \sum_{n=1}^{N} (x[n] - A) \right)^2 \right\}$$

- Since the noise is AWGN, $w[n]$ are uncorrelated and the cross terms becomes zero

$$I(\theta) = \frac{1}{\sigma_x^4} \sum_{n=1}^{N} E\left\{ (x[n] - A)^2 \right\} = \frac{1}{\sigma_x^4} N\sigma_x^2 = \frac{N}{\sigma_x^2}$$

- Hence, the Cramér-Rao lower bound is (again) $\mathrm{Var}(\hat{A}) \geq 1/I(\theta) = \sigma_x^2/N$

General CRLB for signals in WGN (1)

Since it is common to assume some signal in AWGN, it is fruitful to derive the generalised CRLB for this case

- Assume some signal dependent of a parameter in WGN

$$x[n] = s[n; \theta] + w[n], \qquad n = 1, 2, \ldots, N$$

- The likelihood function is

$$f_x(\mathbf{x}; \theta) = \frac{1}{(2\pi\sigma_x^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_x^2} \sum_{n=1}^{N} (x[n] - s[n; \theta])^2 \right)$$

- The derivative of the log likelihood function is

$$\frac{\partial \ln f_x(\mathbf{x}; \theta)}{\partial \theta} = \frac{1}{\sigma_x^2} \sum_{n=1}^{N} (x[n] - s[n; \theta]) \frac{\partial s[n; \theta]}{\partial \theta}$$

remember the chain rule $(f(g(x)))' = f'(g(x))g'(x)$

General CRLB for signals in WGN (2)

The second derivative becomes

$$\begin{aligned}
\frac{\partial}{\partial \theta} \left\{ \frac{1}{\sigma_x^2} \sum_{n=1}^{N} \left(x[n] \frac{\partial s[n; \theta]}{\partial \theta} - s[n; \theta] \frac{\partial s[n; \theta]}{\partial \theta} \right) \right\} \\
= \frac{1}{\sigma_x^2} \sum_{n=1}^{N} \left(x[n] \frac{\partial^2 s[n; \theta]}{\partial \theta^2} - s[n; \theta] \frac{\partial^2 s[n; \theta]}{\partial \theta^2} - \left(\frac{\partial s[n; \theta]}{\partial \theta} \right)^2 \right) \\
= \frac{1}{\sigma_x^2} \sum_{n=1}^{N} \left((x[n] - s[n; \theta]) \frac{\partial^2 s[n; \theta]}{\partial \theta^2} - \left(\frac{\partial s[n; \theta]}{\partial \theta} \right)^2 \right)
\end{aligned}$$

remember the product rule $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$

General CRLB for signals in WGN (3)

Finally, taking the expectation of the second derivative

$$\begin{aligned}
E\left\{ \frac{\partial^2 \ln f_x(\mathbf{x}; \theta)}{\partial \theta^2} \right\} &= E\left\{ \frac{1}{\sigma_x^2} \sum_{n=1}^{N} \left((x[n] - s[n; \theta]) \frac{\partial^2 s[n; \theta]}{\partial \theta^2} - \left(\frac{\partial s[n; \theta]}{\partial \theta} \right)^2 \right) \right\} \\
&= \frac{1}{\sigma_x^2} \sum_{n=1}^{N} \left(E\left\{ \underbrace{(x[n] - s[n; \theta])}_{E\{x[n]\} = s[n; \theta]} \frac{\partial^2 s[n; \theta]}{\partial \theta^2} \right\} - E\left\{ \left(\frac{\partial s[n; \theta]}{\partial \theta} \right)^2 \right\} \right) \\
&= -\frac{1}{\sigma_x^2} \sum_{n=1}^{N} E\left\{ \left(\frac{\partial s[n; \theta]}{\partial \theta} \right)^2 \right\}
\end{aligned}$$

General CRLB for signals in WGN (4)

- The expectation of the second derivative of the log likelihood function then becomes
$$E\left\{\frac{\partial^2 \ln f_X(\mathbf{x};\theta)}{\partial\theta^2}\right\} = -\frac{1}{\sigma_x^2}\sum_{n=1}^{N}\left(\frac{\partial s[n;\theta]}{\partial\theta}\right)^2$$
- The Cramér-Rao lower bound (or the lower bound on variance becomes)
$$\mathrm{Var}(\hat{\theta}) \geq \frac{\sigma_x^2}{\sum_{n=1}^{N}\left(\frac{\partial s[n;\theta]}{\partial\theta}\right)^2}$$
- Note the signal dependence. Signals that change rapidly will result in accurate estimators.

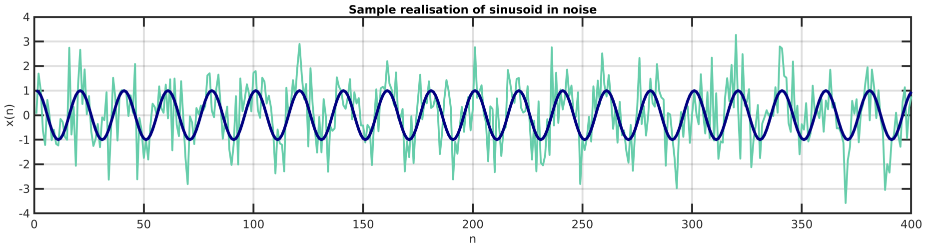
CRLB for a constant value in AWGN - again again

OK. Let's check our new approach on our classic example of a constant value in AWGN

- The model
$$x[n] = A + w[n]$$
- This gives the rather simple expression
$$s[n;\theta] = \theta \quad \Rightarrow \quad \frac{\partial s[n;\theta]}{\partial\theta} = 1$$
- Hence, the CRLB becomes
$$\mathrm{Var}(\hat{\theta}) \geq= \frac{\sigma_x^2}{\sum_{n=1}^{N}\left(\frac{\partial s[n;\theta]}{\partial\theta}\right)^2} = \frac{\sigma_x^2}{N}$$

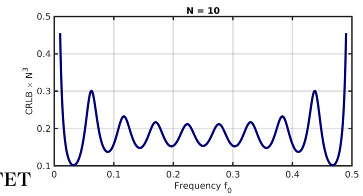
Another example: Frequency estimation in WGN

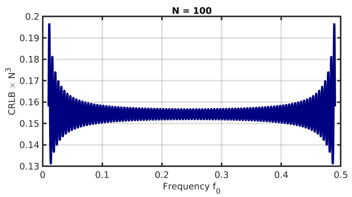
- Assume again a signal dependent of a parameter in WGN
$$x[n] = s[n;\theta] + w[n], \qquad n = 1, 2, \ldots, N$$
where
$$s[n;\theta] = A\cos(2\pi n f_0 + \phi), \qquad 0 < f_0 < 1/2$$
with known amplitude A, known phase ϕ and unknown frequency $f_0 = \theta$



Another example: Frequency estimation in WGN (2)

- Simple calculus gives us
$$\frac{\partial s[n;f_0]}{\partial f_0} = -A2\pi n\sin(2\pi n f_0 + \phi)$$
- Using the general CRLB for a signal in AWGN, we get
$$\mathrm{Var}(\hat{f}_0) \geq \frac{\sigma_x^2}{\sum_{n=1}^{N}\left(\frac{\partial s[n;f_0]}{\partial f_0}\right)^2} = \frac{\sigma_x^2}{A^2\sum_{n=1}^{N}[2\pi n\sin(2\pi n f_0 + \phi)]^2}$$





Cramér-Rao lower bound summarised

- The CRLB is the theoretical bound on the variance any for any unbiased estimator
- It is found by calculating the negative inverse of the expectation of the second derivative of the log likelihood function
- Physically, this can be interpreted as the sharpness (or curvature) of the peak of the likelihood function
- If an unbiased estimator achieves a variance equal to the lower bound, it is said to be an *efficient estimator*
- It is then also the MVU estimator. Not vice versa.
- The CRLB may not exist.

Cramér-Rao lower bound extensions

There are several extensions not mentioned here

- The CRLB is possible to extend to complex random sequences
- The CRLB can be written in a similar manner for estimation of multiple parameters (vector notation)
- In some cases where the theoretical derivation of the CRLB is difficult, it might be possible to calculate an asymptotic CRLB.

These extensions are not too difficult to understand, but the mathematics are slightly more tricky

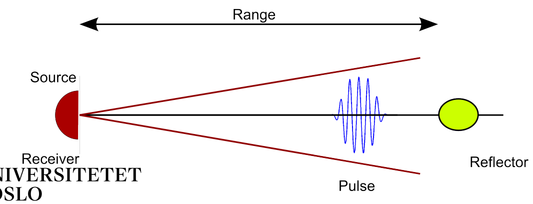
The extensions are well described in Kay's book from 1993.

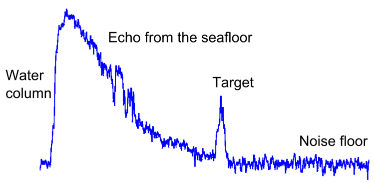
Time delay estimation in sonar (1)

- Consider a transmitted pulse that is reflected by an object in unknown distance
- The range r to the target is equivalent to the two-way travel time of the acoustic pulse

  $$\tau_0 = 2r/c$$

  where c is the sound velocity
- Challenge: given the sonar settings, what is the theoretical lower bound on time delay estimation (equivalent to range estimation)





Time delay estimation in sonar (2)

We describe the time delay estimation problem mathematically as follows

- Consider a transmit pulse $s(t)$ and a received time series $s(t-\tau_0)$ where τ_0 is the wanted delay
- Our signal model becomes

  $$x(t) = s(t-\tau_0) + w(t)$$

  where $w(t)$ is bandlimited additive Gaussian noise
- The transmit signal is assumed to be non-zero during a time interval of T_s (the pulse length)

Time delay estimation in sonar (3)

- We choose the noise bandwidth to be fully covered by the sampling system such that the sampled version of the signal appears to be white
$$x[t] = s(n\Delta - \tau_0) + w[t] \qquad w[t] \text{ WGN}$$
Δ is the sampling interval
- The signal is only non-zero during the pulse length
$$x[n] = \begin{cases} w[n] & 0 \leq n \leq n_0 - 1 \\ s(n\Delta - \tau_0) + w[t] & n_0 \leq n \leq n_0 + M - 1 \\ w[n] & n_0 + M \leq n \leq N - 1 \end{cases}$$
where the received echo has M non-zero samples starting at n_0 and the discrete time delay is given by $\tau_0 = n_0\Delta$

Time delay estimation in sonar (4)

We now apply the general CRLB for signals in WGN

- Replace θ with the desired parameter τ_0

$$\mathrm{Var}(\hat{\tau}_0) \geq \frac{\sigma_x^2}{\sum_{n=0}^{N-1}\left(\frac{\partial s[n;\tau_0]}{\partial \tau_0}\right)^2} = \frac{\sigma_x^2}{\sum_{n=n_0}^{n_0+M-1}\left(\frac{\partial s(n\Delta-\tau_0)}{\partial \tau_0}\right)^2}$$
$$= \frac{\sigma_x^2}{\sum_{n=n_0}^{n_0+M-1}\left(\left.\frac{\partial s(t)}{\partial t}\right|_{t=n\Delta-\tau_0}\right)^2} = \frac{\sigma_x^2}{\sum_{n=0}^{M-1}\left(\left.\frac{\partial s(t)}{\partial t}\right|_{t=n\Delta}\right)^2}$$

- Note that $\tau_0 = n_0\Delta$

Time delay estimation in sonar (5)

- To continue, we assume that the sampling interval is small such that the sum can be replaced by an integral
$$\sum_{n=0}^{M-1}\left(\left.\frac{\partial s(t)}{\partial t}\right|_{t=n\Delta}\right)^2 \approx \frac{1}{\Delta}\int_0^{T_s}\left(\frac{\partial s(t)}{\partial t}\right)^2 dt$$
- Parseval's theorem
$$\int_{-\infty}^{\infty}|s(t)|^2 dt = \int_{-\infty}^{\infty}|S(f)|^2 df$$
- Properties of the Fourier Transform : Time derivative
$$\mathcal{F}\left\{\frac{d}{dt}s(t)\right\} = j2\pi f S(f) \quad \Rightarrow \quad \mathcal{F}\left\{\left(\frac{\partial s(t)}{\partial t}\right)^2\right\} = (2\pi f)^2|S(f)|^2$$

Time delay estimation in sonar (6)

- The *RMS bandwidth* (not to be confused with signal bandwidth) is defined as
$$\beta_{rms}^2 \equiv \frac{\int_{-\infty}^{\infty}(2\pi f)^2|S(f)|^2 df}{\int_{-\infty}^{\infty}|S(f)|^2 df} = \frac{\int_0^{T_s}\left(\frac{\partial s(t)}{\partial t}\right)^2 dt}{\int_0^{T_s} s^2(t)dt}$$
- The signal energy is
$$\mathcal{E} = \int_0^{T_s} s^2(t)dt$$

Time delay estimation in sonar (7)

- Using $\mathcal{E}$ and β_{rms}^2, the lower bound becomes
$$\mathrm{Var}(\hat{\tau}_0) \geq \frac{\sigma_x^2}{\frac{1}{\Delta}\int_0^{T_s}\left(\frac{\partial s(t)}{\partial t}\right)^2 dt} = \frac{\Delta\sigma_x^2}{\mathcal{E}\beta_{rms}^2}$$
- Now, $1/\Delta = 2B$ is the sampling frequency
- The variance of the noise is $\sigma_x^2 = N_0 B$ where N_0 is the noise level
- $\mathcal{E}/(N_0/2)$ is some kind of Signal to Noise Ratio (SNR), we refer to as SNR_E
- Then the CRLB becomes
$$\mathrm{Var}(\hat{\tau}_0) \geq \frac{1}{\mathrm{SNR}_E \times \beta_{rms}^2}$$

Time delay estimation in sonar (8)

- Almost there. SNR_E is not the usual SNR. Let's convert.
- Signal power:
$$P_s = \frac{1}{T_s}\int_0^{T_s} s^2(t)dt = \frac{\mathcal{E}}{T_s}$$
- Noise power:
$$P_n = N_0 B$$
- Which gives
$$\mathrm{Var}(\hat{\tau}_0) \geq \frac{1}{\mathrm{SNR}_E \times \beta_{rms}^2} = \frac{N_0/2}{\mathcal{E} \times \beta_{rms}^2} = \frac{P_n/B}{2P_sT_s \times \beta_{rms}^2} = \frac{1}{2BT_s\,\mathrm{SNR} \times \beta_{rms}^2}$$
- where $\mathrm{SNR} = P_s/P_n$ is the usual SNR

Time delay estimation in sonar (9)

- For flat spectrum $S(f) = S_0$ and signal frequency $[f_1, f_2] = f_0 + [-B/2, B/2]$

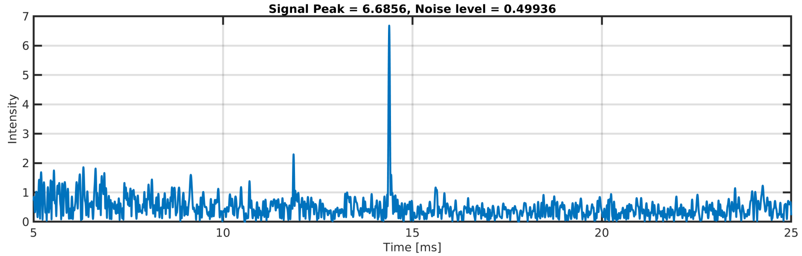
$$\beta_{rms}^2 = \frac{\int_{-\infty}^{\infty}(2\pi f)^2|S(f)|^2 df}{\int_{-\infty}^{\infty}|S(f)|^2 df} = \frac{\int_{f_1}^{f_2}(2\pi f)^2 S_0^2 df}{\int_{f_1}^{f_2} S_0^2 df}$$
$$= (2\pi)^2\frac{S_0^2\left(f_2^3 - f_1^3\right)/3}{S_0^2\left(f_2 - f_1\right)} = \frac{(2\pi)^2}{3}\frac{f_2^3 - f_1^3}{f_2 - f_1} = (2\pi)^2\left(f_0^2 + \frac{B^2}{12}\right)$$

- Or approximately $\beta_{rms}^2 \approx (2\pi)^2 f_0^2$ for $B < f$

Time delay estimation in sonar (10)

- We then obtain
$$\mathrm{Var}(\hat{\tau}_0) \geq \frac{1}{2BT_s\mathrm{SNR} \times \beta_{rms}^2} \approx \frac{1}{2BT_s\mathrm{SNR} \times (2\pi)^2 f_0^2}$$
- The Cramér-Rao lower bound for time delay estimation decreases with increasing SNR, increasing frequency f_0 and increasing number of independent samples BT_s

Time delay estimation in sonar (11): Example



- The signal level is approximately 7 (around the peak) and the noise level is approximately 0.5, giving a SNR of 14.
- The center frequency 100 kHz $\Longrightarrow$ RMS bandwidth $\approx 6.28 \times 10^5$
- The time-bandwidth product $BT_s \approx 1$ after pulse compression

Time delay estimation in sonar (12): Example

- This gives

$$\mathrm{Var}(\hat{\tau}_0) \geq \frac{1}{2BT_s\,\mathrm{SNR} \times (2\pi)^2 f_0^2} = \frac{1}{28 \times 4 \times 10^{11}}\ \mathrm{s}^2$$

- The theoretical lower bound for the standard deviation (the square root of the CRLB) is more understandable

$$\mathrm{std}(\hat{\tau}_0) = \sqrt{\mathrm{Var}(\hat{\tau}_0)} \geq 0.3\,\mu\mathrm{s}$$

- This equals 0.03 wave periods at the carrier frequency

Contents

Least Squares Estimation

- All the previous cases we have studied requires a probabilistic model. We have needed $f_x(\mathbf{x};\theta)$.
- For cases with signal in noise, we actually required a signal model and a noise model
- Least squares is not statistically based.
  - It does not require a statistical model
  - It does, however, require a deterministic signal model
- This approach is not new.
  In 1795 Carl Friedrich Gauss used this method to estimate planetary motions.
- Also known as *regression* in statistical analysis

Carl Friedrich Gauss 1777-1855
German mathematician
From Wikipedia

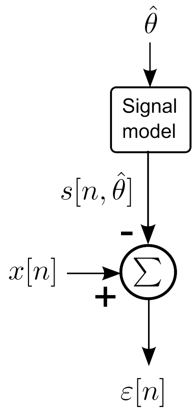
Least Squares Estimation

- We define the least squares cost function as

$$J(\theta) = \sum_{n=1}^{N} \varepsilon^2[n] = \sum_{n=1}^{N} (x[n] - s[n;\theta])^2$$

- We seek to minimise J.
- The *Least Squares Estimator* (LSE) chooses the parameter θ that brings $s[n]$ closest to the observations $x[n]$.
- Can be done by setting the partial derivative to zero

$$\left.\frac{\partial J(\theta)}{\partial \theta}\right|_{\hat{\theta}_{LS}=\theta} = 0$$



Weighted Least Squares Estimation

- The least squares approach is easily extended to a weighted version

$$J(\theta) = \sum_{n=1}^{N} w_n\,(x[n] - s[n;\theta])^2$$

where w_n is some weight chosen to increase or decrease the individual importance of each sample in the observation.
- Typically, this weight is related to data quality or SNR

A word of caution

If the chosen model does not reflect the observations in a proper way, the estimate will be biased. This applies both for the weighted and ordinary LS estimation

Linear Least Squares Estimation

- The *linear least squares* problem is when the parameter observation model is linear.
- Assume that we have p unknown variables to be estimated $\boldsymbol{\theta} = [\theta_1, \theta_2, \ldots, \theta_p]^T$
- The observation model is linear $\mathbf{s} = \mathbf{H}\boldsymbol{\theta}$
- $\mathbf{H}$ is the known $N \times p$ size *observation matrix*
- Example: Fitting a linear line to the data

$$s[n] = An + B \quad \Rightarrow \quad \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ \vdots & \vdots \\ 1 & N-1 \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} A \\ B \end{bmatrix}}_{\boldsymbol{\theta}}$$

Linear Least Squares Estimation

- The cost function for linear least squares becomes

$$J(\boldsymbol{\theta}) = \sum_{n=1}^{N} (x[n] - s[n;\boldsymbol{\theta}])^2 = (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})$$

- By solving

$$\frac{\partial J(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{0}$$

we obtain a closed form of the linear least squares estimator

Linear Least Squares Estimation

- The cost function written out
  $$J(\boldsymbol{\theta}) = \mathbf{x}^T\mathbf{x} - \mathbf{x}^T\mathbf{H}\boldsymbol{\theta} - \boldsymbol{\theta}^T\mathbf{H}^T\mathbf{x} + \boldsymbol{\theta}^T\mathbf{H}^T\mathbf{H}\boldsymbol{\theta}$$
- The gradient
  $$\frac{\partial J(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = -2\mathbf{H}^T\mathbf{x} + 2\mathbf{H}^T\mathbf{H}\boldsymbol{\theta}$$
- Setting equal to zero
  $$\mathbf{H}^T\mathbf{H}\boldsymbol{\theta} = \mathbf{H}^T\mathbf{x}$$
- which gives a closed form of the linear least squares estimator
  $$\hat{\boldsymbol{\theta}}_{LS} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$$

See *Solving Linear Systems of Equations* in the matlab manual

Linear Least Squares Estimation - single parameter case

- Now, consider the single parameter case $s[n] = \theta h[n]$
- The cost function becomes
  $$J(\theta) = \sum_{n=1}^{N} (x[n] - \theta h[n])^2$$
- The gradient
  $$\frac{\partial J(\theta)}{\partial \theta} = -2\sum_{n=1}^{N} (x[n] - \theta h[n])\, h[n]$$
- Setting equal zero and solving for θ gives the LS estimate
  $$\hat{\theta}_{LS} = \sum_{n=1}^{N} x[n]h[n] \bigg/ \sum_{n=1}^{N} h^2[n]$$

Summary of Least Squares Estimation

- Least Squares Estimation (LSE) is essentially to fit a model $s(\theta)$ to a data set x by minimising the squared error
- The sum of the squared error is called the cost function
- LSE does not require any probabilistic information
- LSE does however require a deterministic model
- If the model can be described as a linear combination of the wanted parameters, linear LSE can be applied
- The linear LSE provides a closed form solution
- There are numerous extensions of the least squares approach
- If the model does not represent the observations properly, the estimates will be poorer and probably have bias

Example: Fine time delay estimation

We will now calculate the linear least squares estimator for a simple case of time delay estimation

- Consider the time delay example in the previous section
- Assume that the received echo around the maximum value is a Gauss
  $$x[n] = I_0 \exp(-(t-\tau_0)^2/2\sigma_\tau^2) \quad t = n\Delta$$
- This equation has three unknown parameters we wish to estimate: peak intensity I_0, time delay τ_0, and pulse width σ_τ
- We realise that this model becomes a quadratic function if we take the logarithm
  $$x'[n] = \ln x[n] = \ln I_0 - (t-\tau_0)^2/2\sigma_\tau^2$$
- We can reorganise this equation to fit the form
  $$\theta_0 + \theta_1 t + \theta_2 t^2$$

Example: Fine time delay estimation

- The *observation matrix* becomes then

$$\mathbf{H} = \begin{bmatrix} 1 & t[1] & t[1]^2 \\ 1 & t[2] & t[2]^2 \\ 1 & t[3] & t[3]^2 \\ \vdots & \vdots & \vdots \\ 1 & t[N] & t[N]^2 \end{bmatrix}$$

- The least squares estimator is

$$\hat{\boldsymbol{\theta}}_{LS} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$$

- This is simply `theta = H\x` in matlab
- See *Solving Linear Systems of Equations* in the matlab manual

Example: Fine time delay estimation

- The least squares estimated values

$$\hat{\boldsymbol{\theta}}_{LS} = [\hat{\theta}_0, \hat{\theta}_1, \hat{\theta}_2]^T$$

can be reorganised into the wanted parameters (by simple algebra) as follows

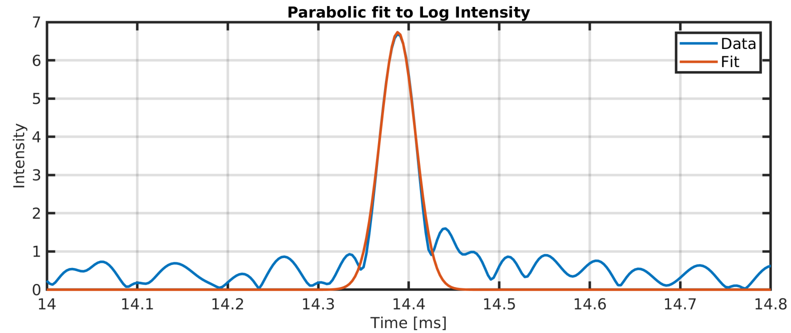
$$\begin{aligned} \hat{\sigma}_\tau^2 &= -\frac{1}{2\hat{\theta}_2} \\ \hat{\tau}_0 &= -\frac{\hat{\theta}_1}{2\hat{\theta}_2} \\ \hat{I}_0 &= \exp\left(\hat{\theta}_0 - \frac{\hat{\theta}_1^2}{4\hat{\theta}_2}\right) \end{aligned}$$

Fine time delay estimation - Algorithm

- Locate the sample delay for the maximum value of the magnitude in the timeseries
- Upsample the timeseries to get a decent number of samples on the peak (a factor of 4 or 8)
- Select the samples around peak that has sample values above the half-value of peak
- Remember to take the logarithm of the data
- Run the linear least squares method to estimate amplitude, time delay and pulse width
- The cost function can be used as a measure for model misfit

Fine time delay estimation - Result

$$\hat{\sigma}_\tau = 0.0194 \text{ ms}$$
$$\hat{\tau}_0 = 14.39 \text{ ms}$$
$$\hat{I}_0 = 6.73$$



Contents

Summary of lecture IV

List of terms from this lecture

- Parameter estimation
- Likelihood function
- Maximum likelihood estimation
- Log likelihood function (LLF)
- Cramér-Rao lower bound (CRLB)
- Fisher Information
- Efficient estimator
- Least Squares Estimator